Communication Complexity of Exact Sampling under Exponential Cost

Spencer Hill

Queen's University, Canada

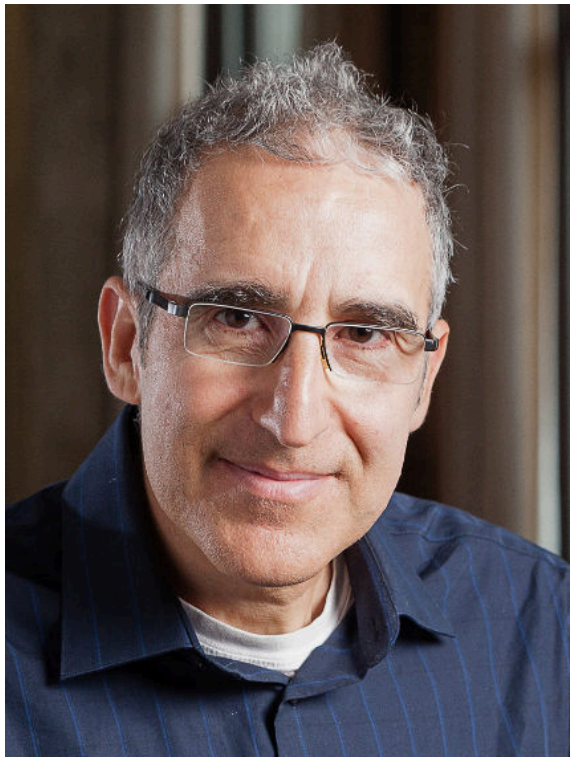
CMS Winter Meeting

December 6, 2025

Joint work with



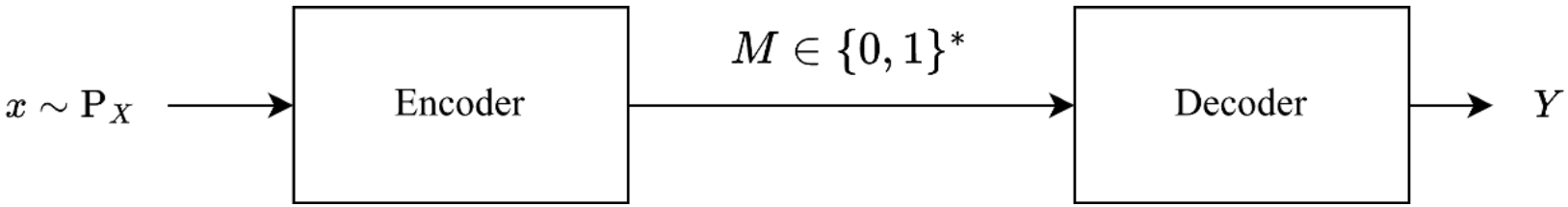
Tamás Linder



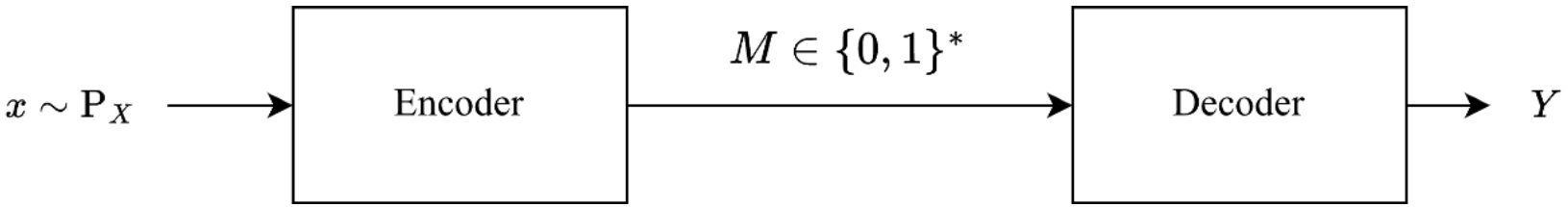
Fady Alajaji

Queen's University

Lossy Compression

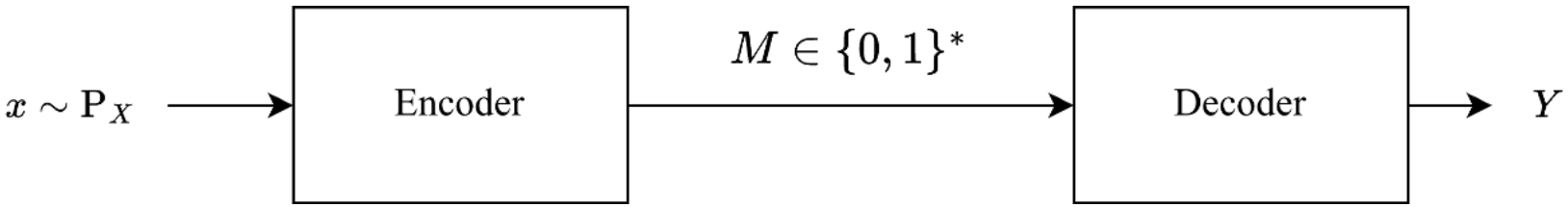


Lossy Compression



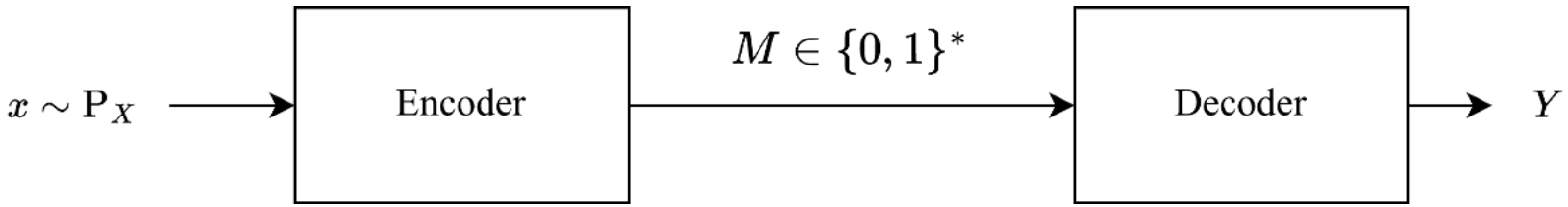
- The encoder encodes X (image) into bits M

Lossy Compression



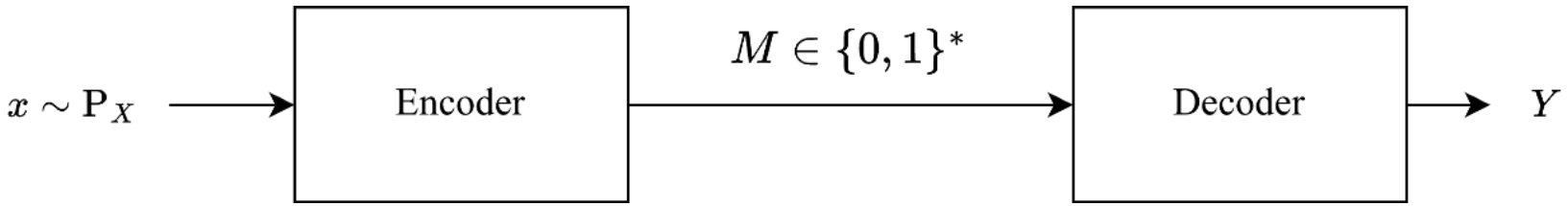
- The encoder encodes X (image) into bits M
- Decoder reconstructs Y "close" to X

Lossy Compression



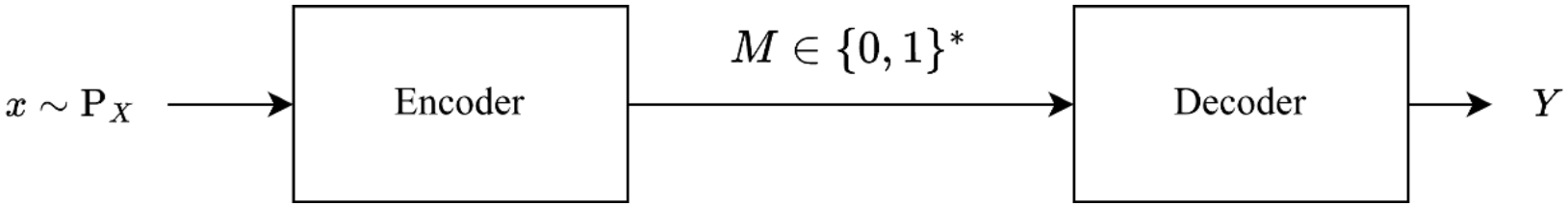
- The encoder encodes X (image) into bits M
- Decoder reconstructs Y "close" to X
- Metrics:
  - Distortion: $D = \mathbb{E}\big[d(X,Y)\big]$

Lossy Compression



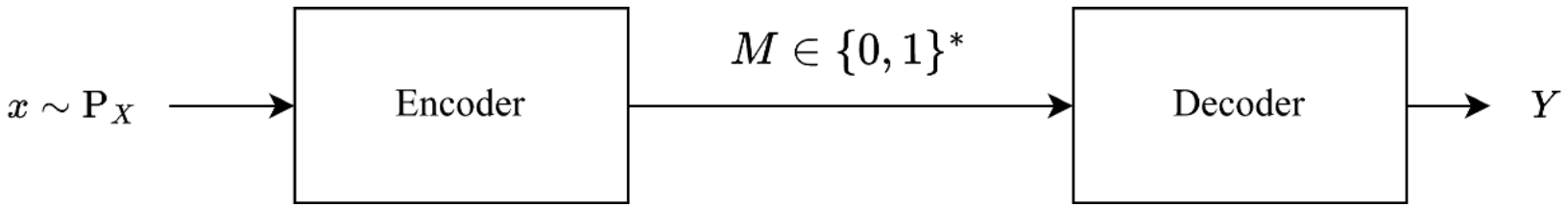
- The encoder encodes X (image) into bits M
- Decoder reconstructs Y "close" to X
- Metrics:
  - Distortion: $D = \mathbb{E}\big[d(X,Y)\big]$
  - Rate: $R = \mathbb{E}|M|$ (expected message length)

The Usual Implementation



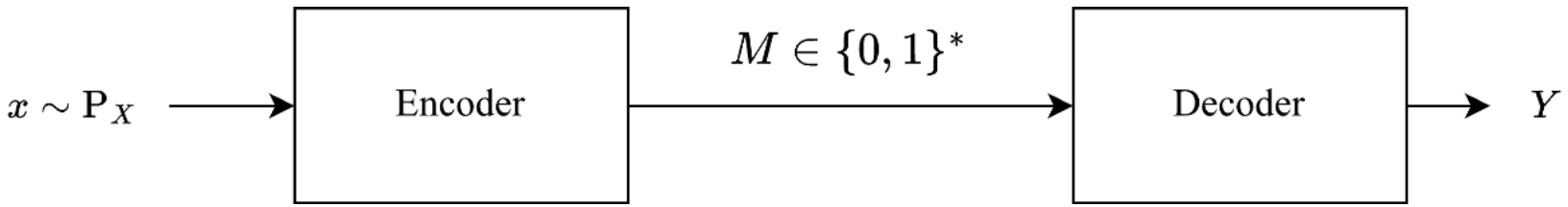
1. Quantizer Q

The Usual Implementation



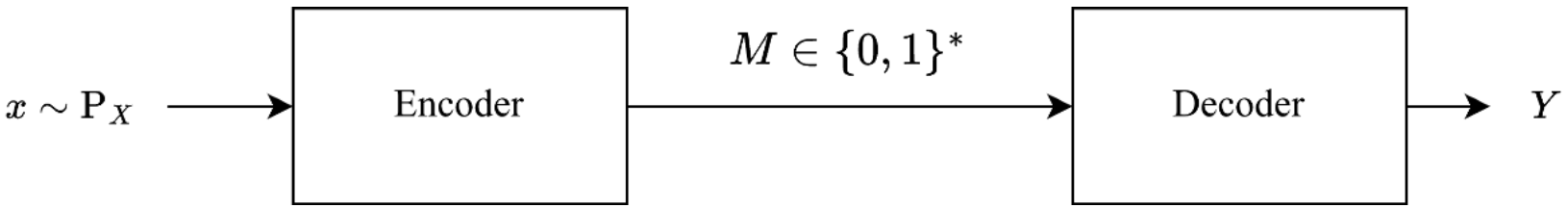
1. Quantizer Q
2. Source distribution $\hat{P} = Q_{\#}P$

The Usual Implementation



1. Quantizer Q
2. Source distribution $\hat{P} = Q_{\#}P$
3. Lossless source code $K_{\hat{P}}$

The Usual Implementation

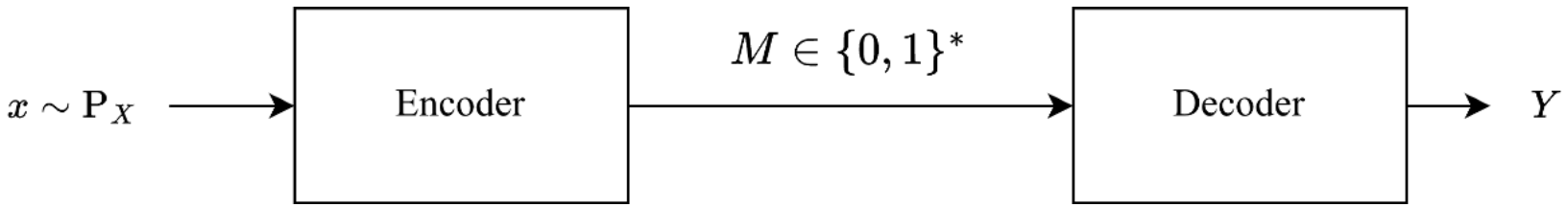


1. Quantizer Q
2. Source distribution $\hat{P} = Q_{\#}P$
3. Lossless source code $K_{\hat{P}}$

The scheme is then:

- $M = K_{\hat{P}} \circ Q$

The Usual Implementation

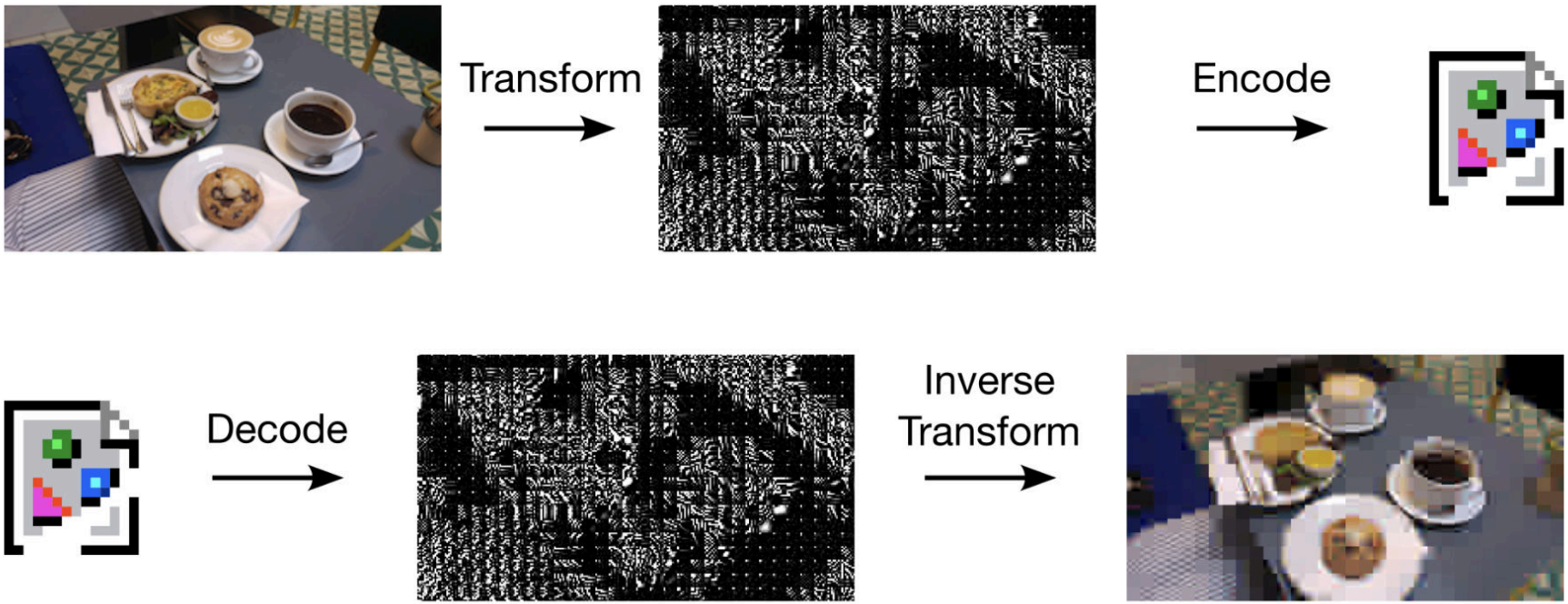


1. Quantizer Q
2. Source distribution $\hat{P} = Q_{\#}P$
3. Lossless source code $K_{\hat{P}}$

The scheme is then:

- $M = K_{\hat{P}} \circ Q$
- $Y = K_{\hat{P}}^{-1}(M)$

Transform Coding (JPEG)



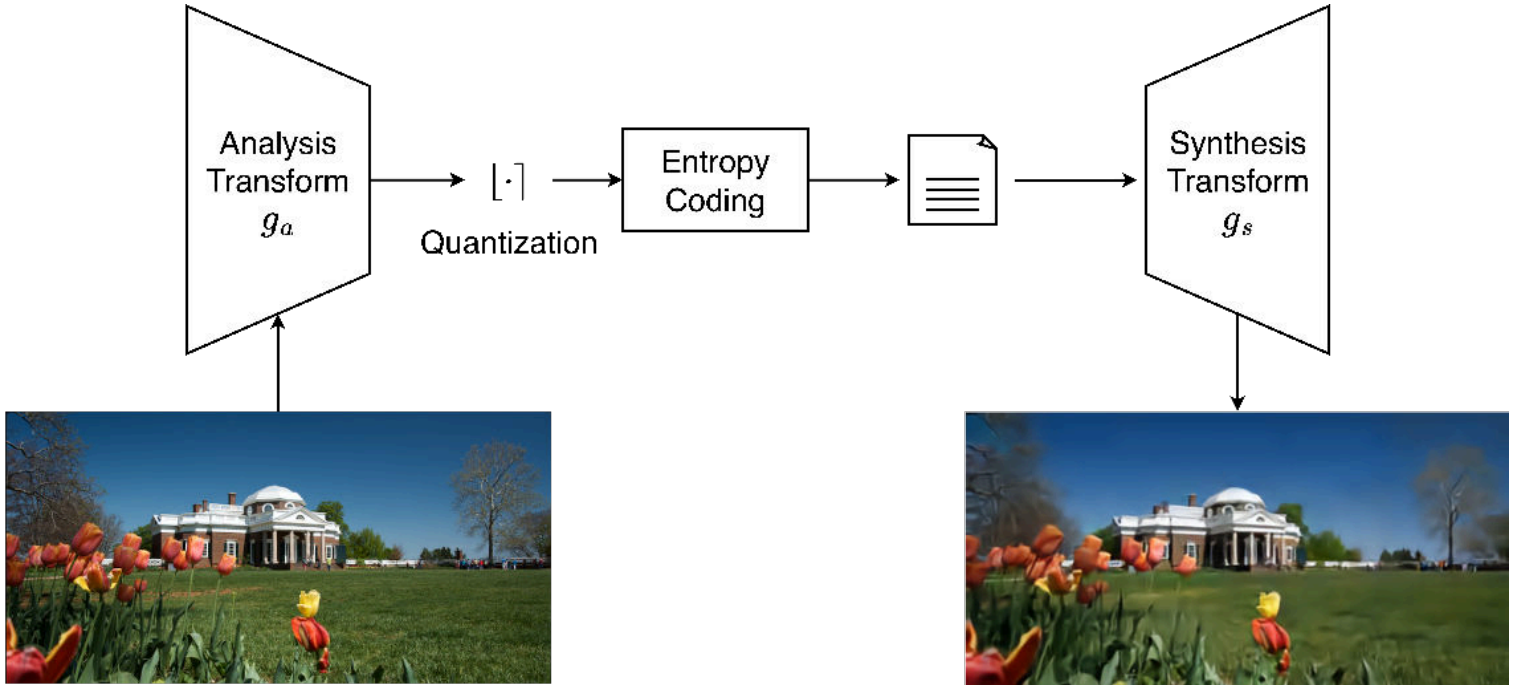
Usual transform: $Q \circ f$; for JPEG f is the discrete cosine transform

Neural Transform Coding

Idea: Learn the transform f using a neural network

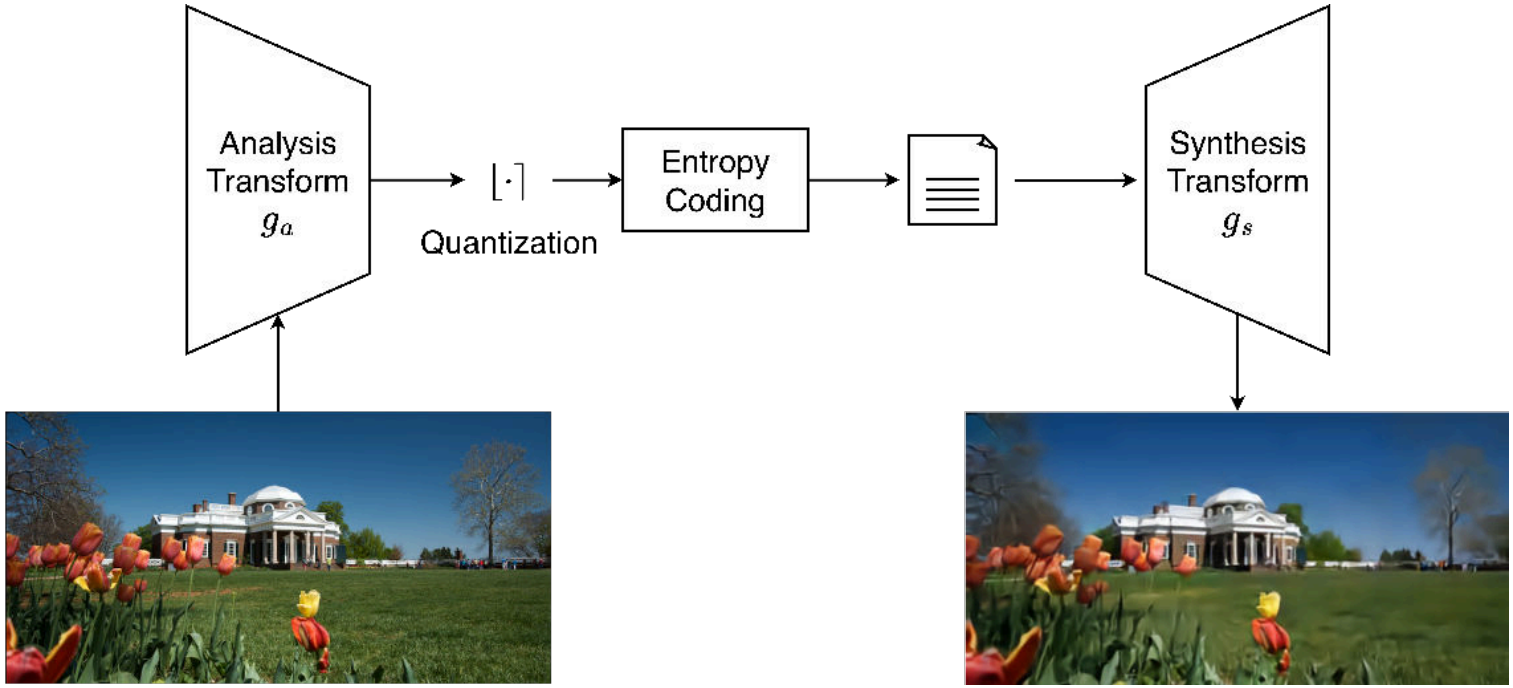
Neural Transform Coding

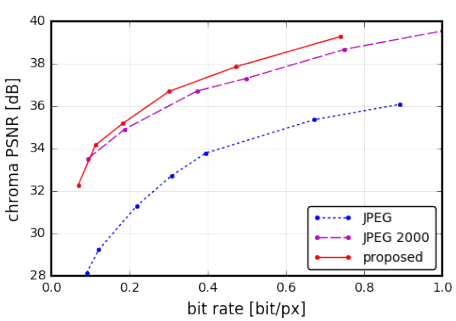
Idea: Learn the transform f using a neural network



Neural Transform Coding

Idea: Learn the transform f using a neural network



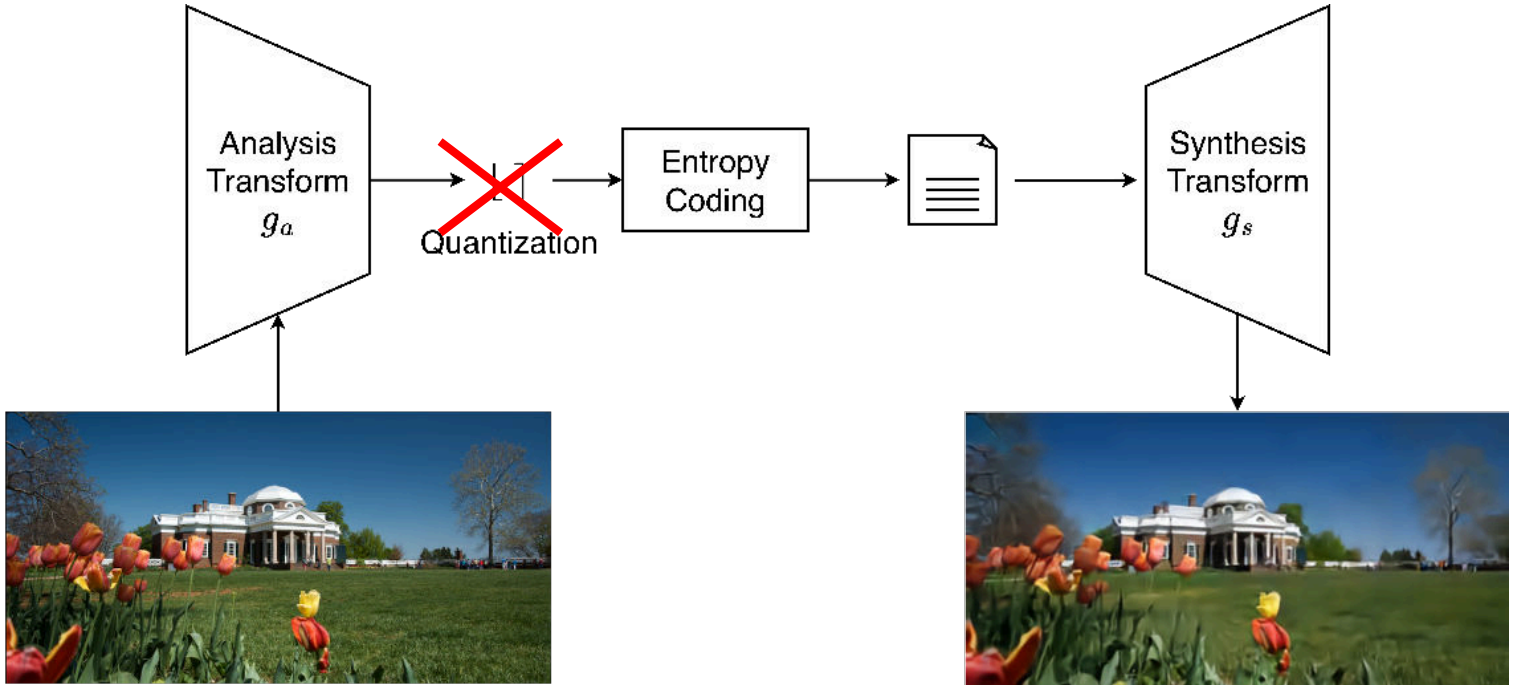


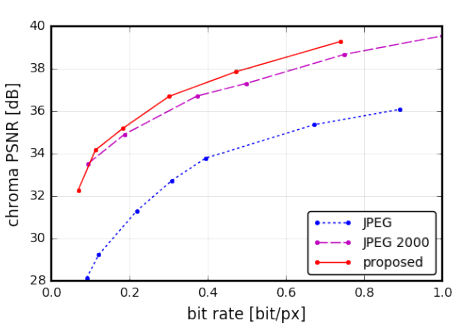


JPEG

Neural Transform Coding

Idea: Learn the transform f using a neural network

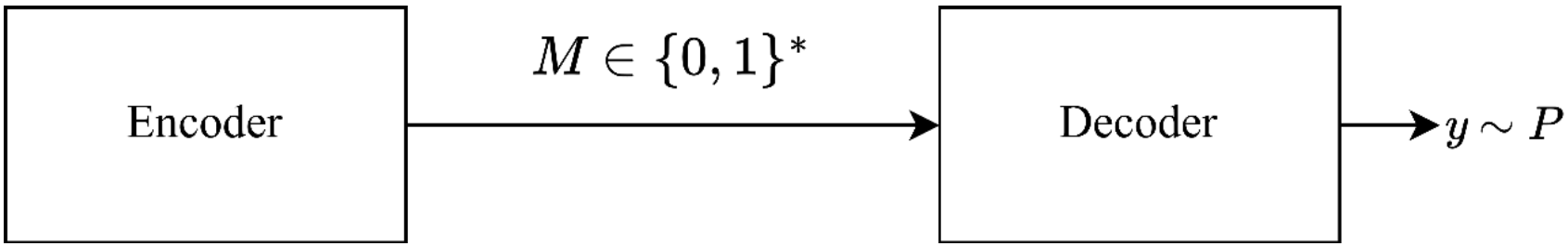






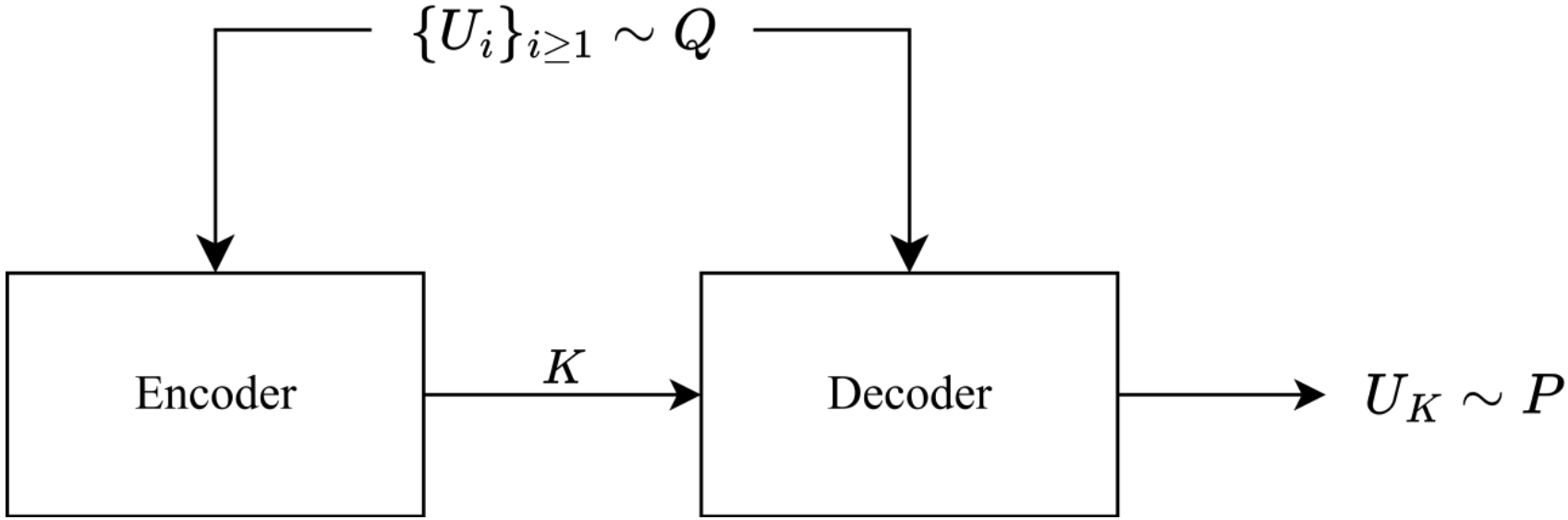
JPEG

Exact Sampling



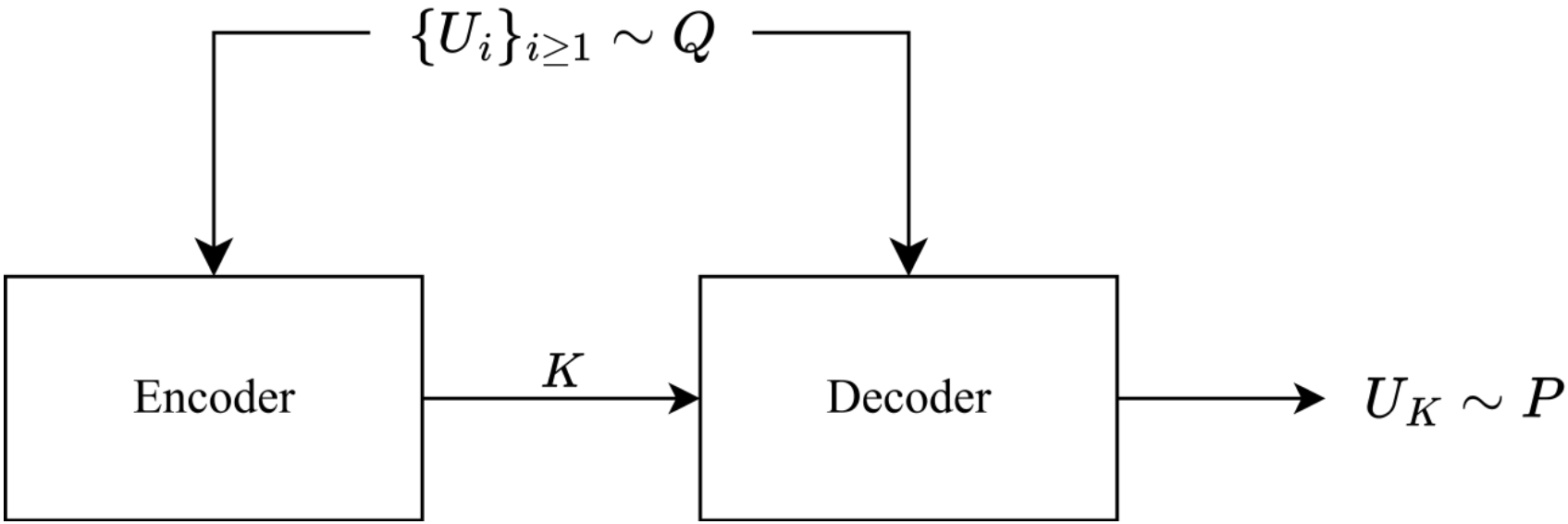
Goal: Send a binary message M so the receiver can draw a sample with distribution P.

Exact Sampling and Channel Simulation



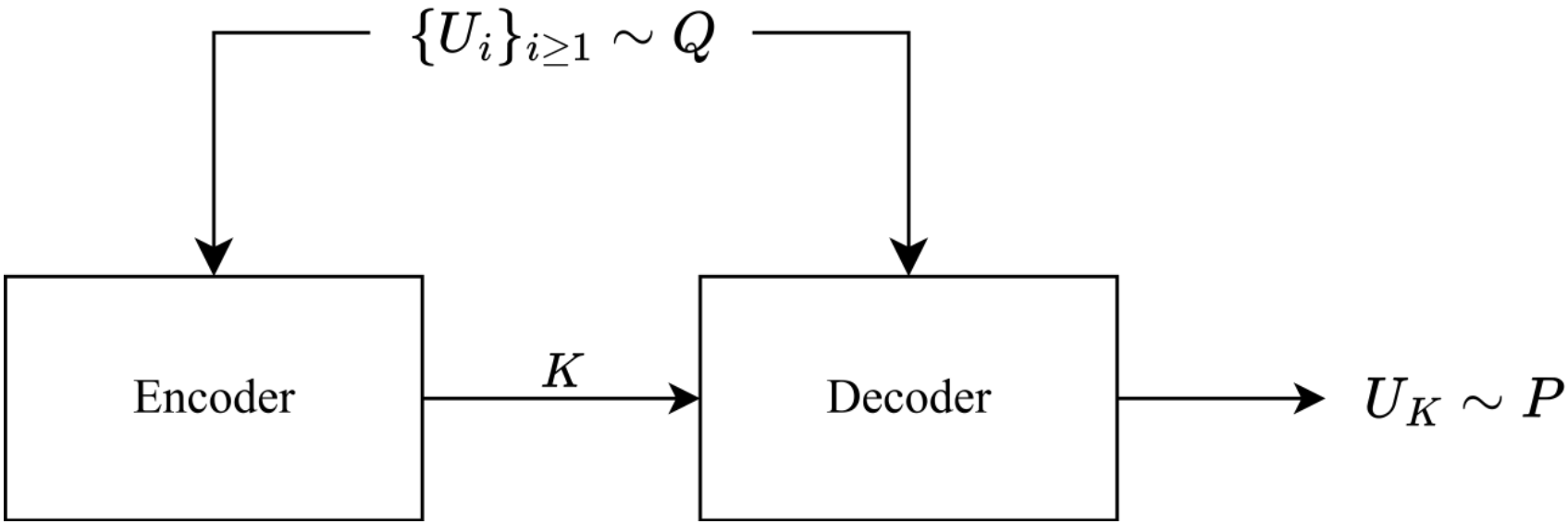
- Shared randomness $\{U_i\}_{i\geq 1} \overset{\text{i.i.d.}}{\sim} Q$

Exact Sampling and Channel Simulation



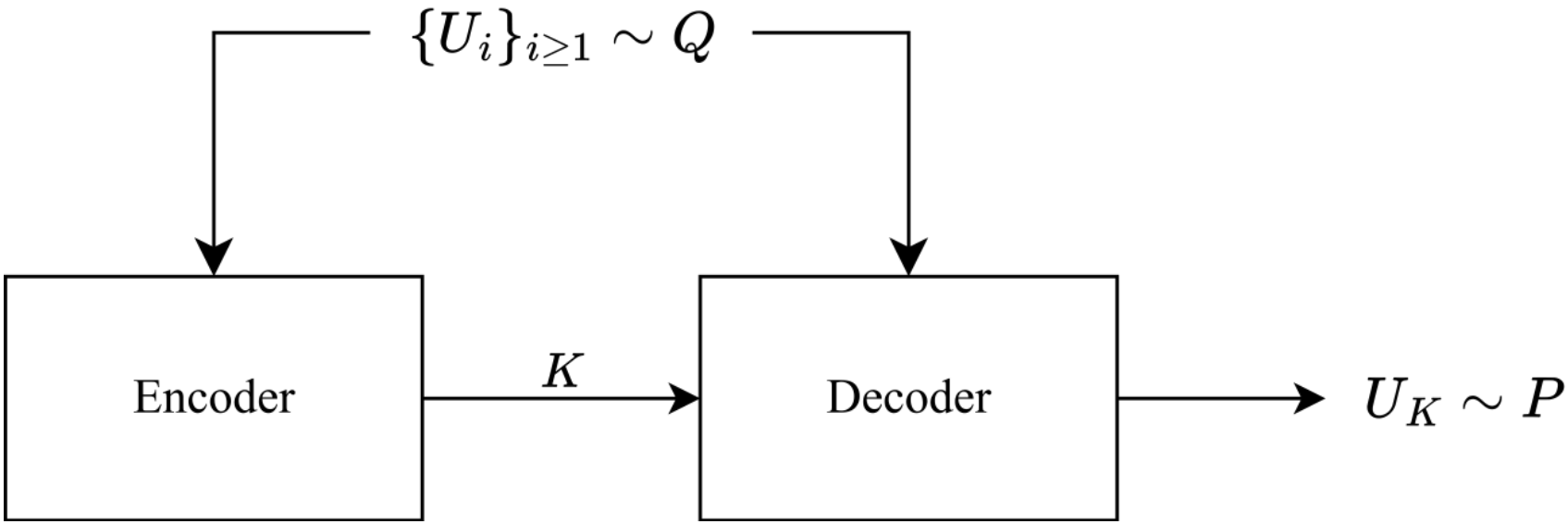
- Shared randomness $\{U_i\}_{i\geq 1} \overset{\text{i.i.d.}}{\sim} Q$
- (Random) index K s.t. $U_K \sim P$.

Exact Sampling and Channel Simulation



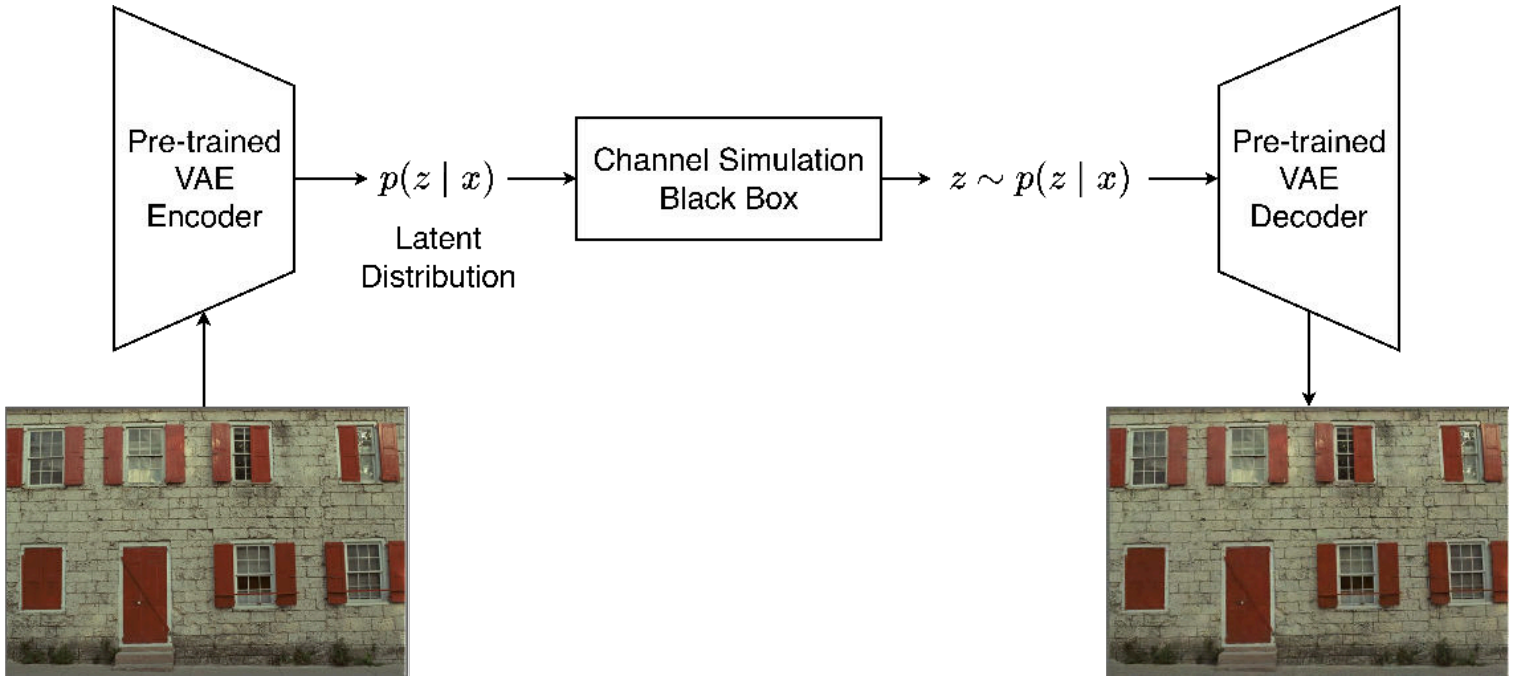
- Shared randomness $\{U_i\}_{i\geq 1} \overset{\text{i.i.d.}}{\sim} Q$
- (Random) index K s.t. $U_K \sim P$.
- Channel simulation: set $Q = \mathrm{P}_Y$, on input $x \sim \mathrm{P}_X$ set $P = \mathrm{P}_{Y|X}(\,\cdot\,|\,x)$; simulates the channel $X \to Y$.

Exact Sampling and Channel Simulation

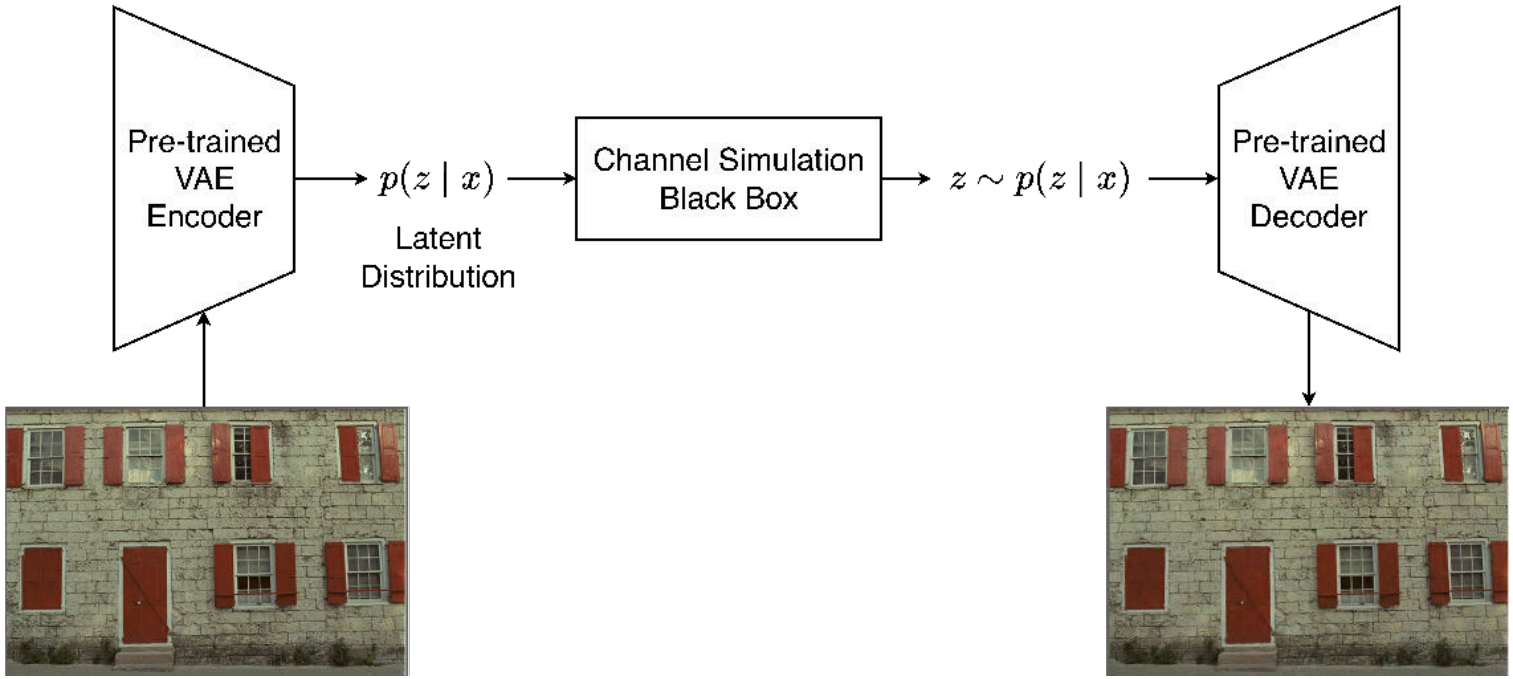


- Shared randomness $\{U_i\}_{i\geq 1} \overset{\text{i.i.d.}}{\sim} Q$
- (Random) index K s.t. $U_K \sim P$.
- Channel simulation: set $Q = \mathrm{P}_Y$, on input $x \sim \mathrm{P}_X$ set $P = \mathrm{P}_{Y|X}(\,\cdot\,|\,x)$; simulates the channel $X \to Y$.
- Can achieve $\mathbb{E}|M| \approx D(P||Q)$ bits, for channel simulation this is $\approx E_X[D(\mathrm{P}_{Y|X}(\,\cdot\,|\,x)||\mathrm{P}_Y)] = I(X;Y)$ bits **(Optimal)**.

Neural Compression with Channel Simulation



Neural Compression with Channel Simulation



Fully differentiable end-to-end system!

Other Applications

- Realizing the optimal compression channel in lossy source coding

Other Applications

- Realizing the optimal compression channel in lossy source coding
- Rate-distortion-perception tradeoff

Other Applications

- Realizing the optimal compression channel in lossy source coding
- Rate-distortion-perception tradeoff
- Compression via implicit neural representation

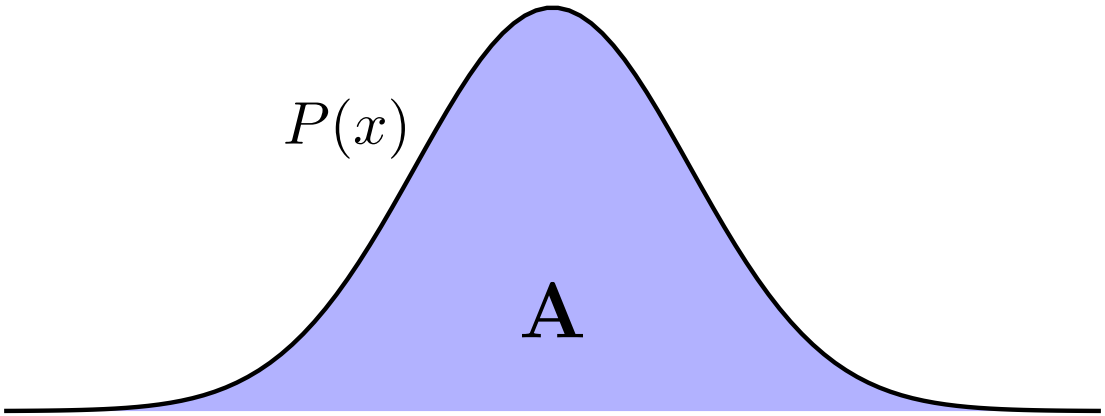
Other Applications

- Realizing the optimal compression channel in lossy source coding
- Rate-distortion-perception tradeoff
- Compression via implicit neural representation
- Local differential privacy

Other Applications

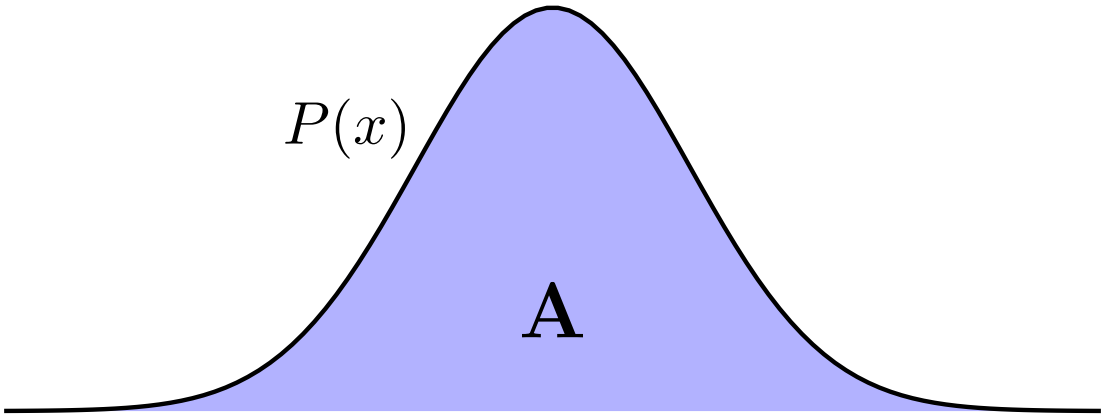
- Realizing the optimal compression channel in lossy source coding
- Rate-distortion-perception tradeoff
- Compression via implicit neural representation
- Local differential privacy
- Federated learning, ...

Some Intuition: Rejection Sampling



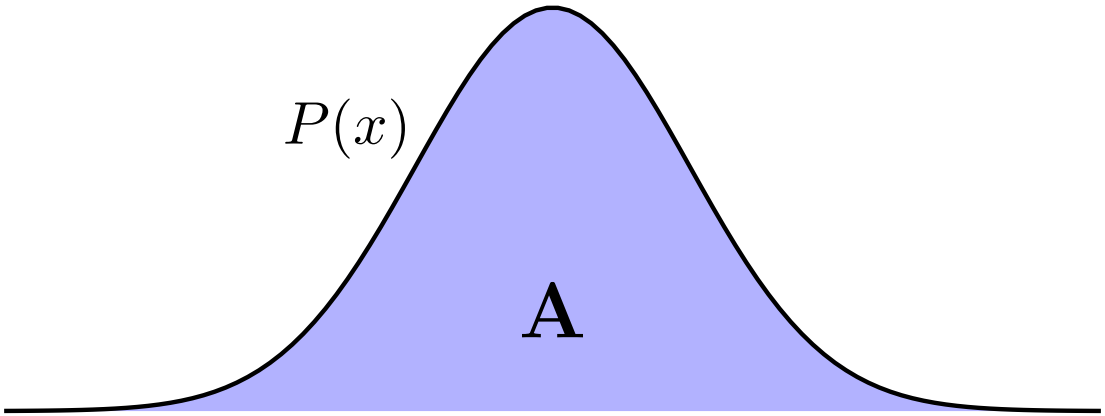
- Fact: $(x, y) \sim \text{Unif}(A) \implies x \sim P$.

Some Intuition: Rejection Sampling



- Fact: $(x, y) \sim \text{Unif}(A) \implies x \sim P$.
- Algorithmic Interpretation: If samples U_k are drawn uniformly, accept sample U_k with probability $P(U_k)$.

Some Intuition: Rejection Sampling



- Fact: $(x, y) \sim \text{Unif}(A) \implies x \sim P$.
- Algorithmic Interpretation: If samples U_k are drawn uniformly, accept sample U_k with probability $P(U_k)$.
- General intuition (nonuniform Q): accept U_k with probability $\propto \frac{\mathrm{d}P}{\mathrm{d}Q}(U_k)$.

State-of-the-Art Sampling Algorithms

- **Greedy Rejection Sampling:** Accept/reject each U_k sequentially, maximizing the acceptance probability each stage under the condition that the scheme is exact.

State-of-the-Art Sampling Algorithms

- **Greedy Rejection Sampling:** Accept/reject each U_k sequentially, maximizing the acceptance probability each stage under the condition that the scheme is exact.

$$D(P||Q) \leq \mathbb{E}|M| \leq D(P||Q) + \log_2(D(P||Q) + 1) + 4$$

State-of-the-Art Sampling Algorithms

- **Greedy Rejection Sampling:** Accept/reject each U_k sequentially, maximizing the acceptance probability each stage under the condition that the scheme is exact.

$$D(P||Q) \leq \mathbb{E}|M| \leq D(P||Q) + \log_2(D(P||Q) + 1) + 4$$

- **Poisson Functional Representation:** For $\{T_i\}_{i\geq 1}$ a rate-one Poisson process, choose $K = \arg\min_{i\geq 1} \frac{T_i}{\frac{\mathrm{d}P}{\mathrm{d}Q}(U_i)}$.

State-of-the-Art Sampling Algorithms

- **Greedy Rejection Sampling:** Accept/reject each U_k sequentially, maximizing the acceptance probability each stage under the condition that the scheme is exact.

$$D(P||Q) \leq \mathbb{E}|M| \leq D(P||Q) + \log_2(D(P||Q) + 1) + 4$$

- **Poisson Functional Representation:** For $\{T_i\}_{i\geq 1}$ a rate-one Poisson process, choose $K = \arg\min_{i\geq 1} \frac{T_i}{\frac{\mathrm{d}P}{\mathrm{d}Q}(U_i)}$.

$$D(P||Q) \leq \mathbb{E}|M| \leq D(P||Q) + \log_2(D(P||Q) + 2) + 3$$

Our Setup: Exponential Cost and Rényi's entropy

- The previous results are for the expected message length (number of bits) $\mathbb{E}|M|$.

Our Setup: Exponential Cost and Rényi's entropy

- The previous results are for the expected message length (number of bits) $\mathbb{E}|M|$.
- What are the fundamental limits of exact sampling and channel simulation under a cost which is *exponential* in the message lengths? Can these limits be (almost) achieved by existing algorithms?

Some Definitions

For a discrete random variable K with pmf P_K and any $\alpha \in (0,1) \cup (1,\infty)$, the Rényi entropy $H_\alpha(K)$ is

$$H_\alpha(X) = \frac{1}{1-\alpha} \log\left(\sum_{k \in \mathcal{K}} \mathrm{P}_K(k)^\alpha \right).$$

Some Definitions

For a discrete random variable K with pmf P_K and any $\alpha \in (0,1) \cup (1,\infty)$, the Rényi entropy $H_\alpha(K)$ is

$$H_\alpha(X) = \frac{1}{1-\alpha} \log\left(\sum_{k \in \mathcal{K}} \mathrm{P}_K(k)^\alpha \right).$$

For $P \ll Q$ probability distributions and $\alpha \in (0,1) \cup (1,\infty)$, the Rényi divergence $D_\alpha(P||Q)$ is

$$D_\alpha(P||Q) = \frac{1}{\alpha - 1} \log\left(\mathbb{E}_{X \sim Q} \left[\left(\frac{\mathrm{d}P}{\mathrm{d}Q}(X) \right)^\alpha \right] \right).$$

Some Definitions

For a discrete random variable K with pmf P_K and any $\alpha \in (0,1) \cup (1,\infty)$, the Rényi entropy $H_\alpha(K)$ is

$$H_\alpha(X) = \frac{1}{1-\alpha} \log\left(\sum_{k \in \mathcal{K}} \mathrm{P}_K(k)^\alpha \right).$$

For $P \ll Q$ probability distributions and $\alpha \in (0,1) \cup (1,\infty)$, the Rényi divergence $D_\alpha(P||Q)$ is

$$D_\alpha(P||Q) = \frac{1}{\alpha - 1} \log\left(\mathbb{E}_{X \sim Q} \left[\left(\frac{\mathrm{d}P}{\mathrm{d}Q}(X) \right)^\alpha \right] \right).$$

Note that $\lim_{\alpha \to 1} H_\alpha(K) = H(K)$ and $\lim_{\alpha \to 1} D_\alpha(P||Q) = D(P||Q)$.

Campbell Cost $L(t)$

For a binary encoding $M \in \{0,1\}^*$ of K having length $|M|$ and for $t > 0$,

$$L(t) := \frac{1}{t} \log\Big(\mathbb{E}[2^{t|M|}]\Big).$$

Campbell Cost $L(t)$

For a binary encoding $M \in \{0,1\}^*$ of K having length $|M|$ and for $t > 0$,

$$L(t) := \frac{1}{t} \log\Big(\mathbb{E}[2^{t|M|}]\Big).$$

Facts:

$$\lim_{t\to 0} L(t) = \mathbb{E}|M| \qquad \text{and} \qquad \lim_{t\to\infty} L(t) = \max_{\ell\in\mathbb{N}\,:\,\mathbb{P}(|M|=\ell)>0} \ell$$

Campbell Cost $L(t)$

For a binary encoding $M \in \{0,1\}^*$ of K having length $|M|$ and for $t > 0$,

$$L(t) := \frac{1}{t} \log\Big(\mathbb{E}[2^{t|M|}]\Big).$$

Facts:

$$\lim_{t\to 0} L(t) = \mathbb{E}|M| \qquad \text{and} \qquad \lim_{t\to\infty} L(t) = \max_{\ell\in\mathbb{N}\,:\,\mathbb{P}(|M|=\ell)>0} \ell$$

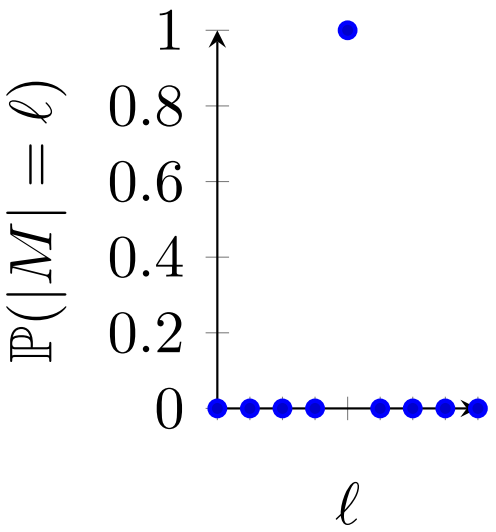
For a random variable K with Rényi entropy $H_\alpha(K)$ encoded optimally into message M, Campbell (1965) showed

$$H_\alpha(K) \le L(t) < H_\alpha(K) + 1$$

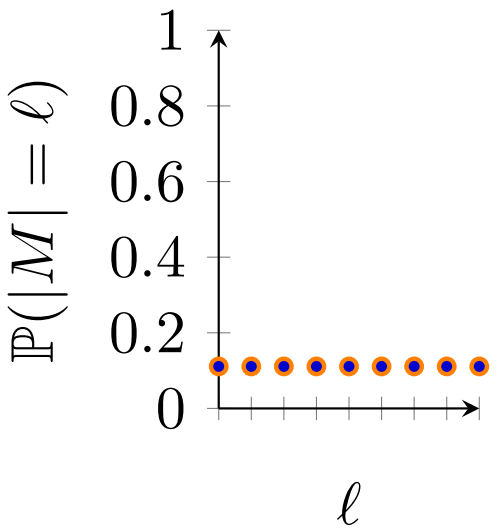
with $\alpha = \frac{1}{1+t}$.

Why Care About $L(t)$?

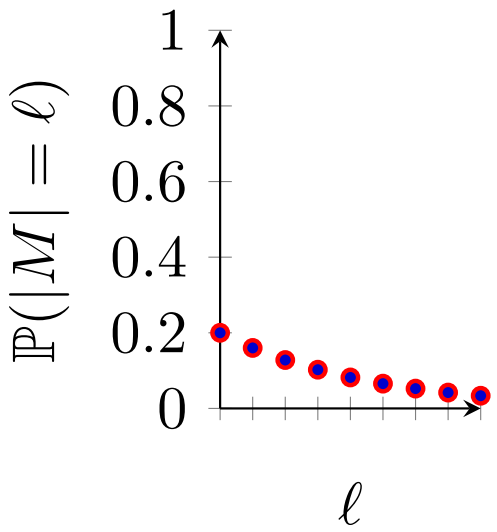
Degenerate at 5



Uniform on $\{1, \ldots, 9\}$

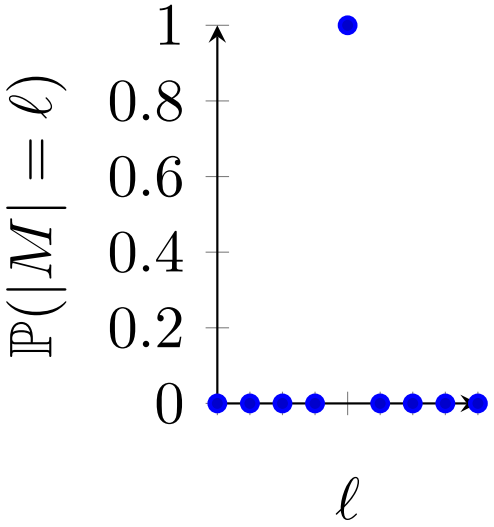


Geometric ($p = 0.2$)

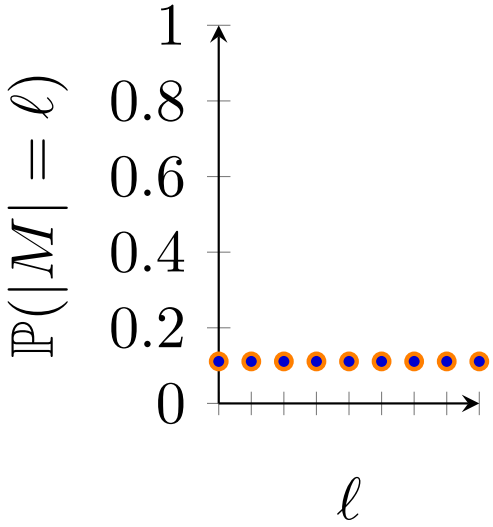


Why Care About $L(t)$?

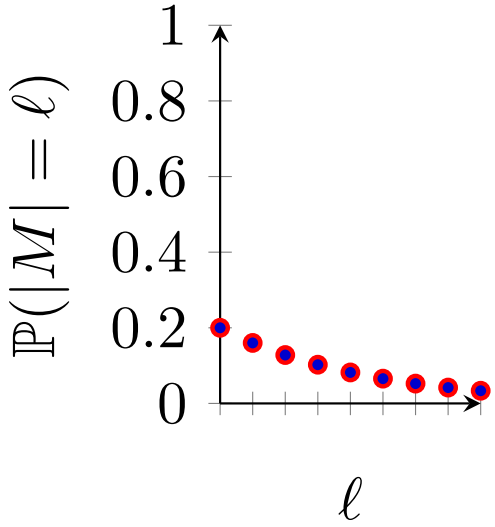
Degenerate at 5



Uniform on $\{1, \ldots, 9\}$



Geometric ($p = 0.2$)



t	**Degenerate** $L(t)$	**Uniform** $L(t)$	**Geometric** $L(t)$
0	5	5	5
0.2	5	5.65	11.83
1	5	7.26	∞
5	5	8.56	∞
∞	5	9	∞

(Almost) Tight Bounds

Theorem 1 For any sampling algorithm and $t > 0$, with $\alpha = \frac{1}{1+t}$,

$$L(t) \geq D_{\frac{1}{\alpha}}(P||Q) + \frac{\alpha}{1-\alpha}\log_2(\alpha) - 1. \tag{1}$$

As $t \to 0$, we recover the lower bound $\mathbb{E}[|M|] \geq D(P||Q) - \frac{1}{\ln(2)} - 1$.

(Almost) Tight Bounds

Theorem 1 For any sampling algorithm and $t > 0$, with $\alpha = \frac{1}{1+t}$,

$$L(t) \geq D_{\frac{1}{\alpha}}(P||Q) + \frac{\alpha}{1-\alpha}\log_2(\alpha) - 1. \tag{1}$$

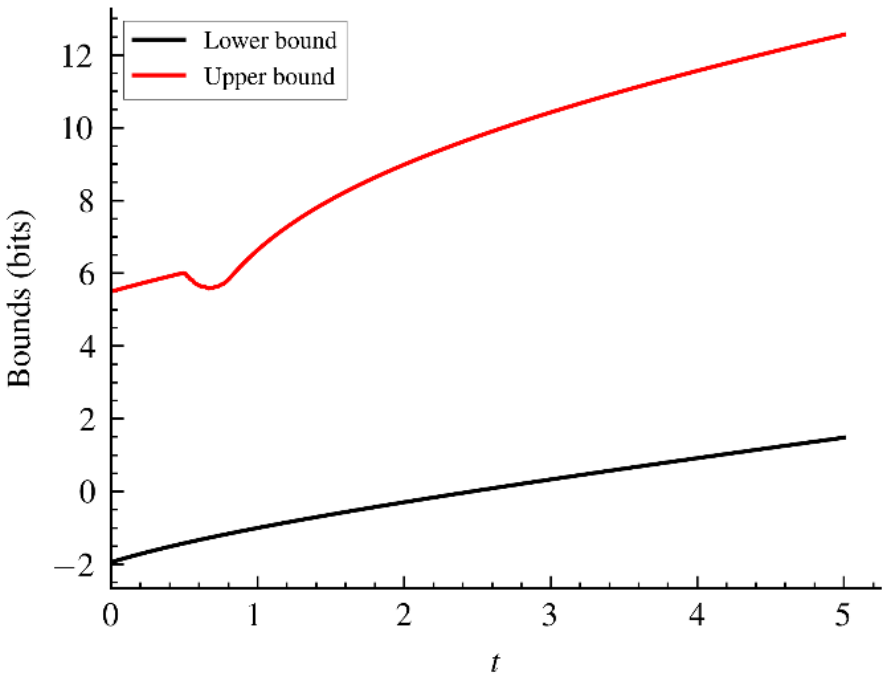
As $t \to 0$, we recover the lower bound $\mathbb{E}[|M|] \geq D(P||Q) - \frac{1}{\ln(2)} - 1$.

Theorem 2 For K chosen using the Poisson functional representation, for any $\epsilon > 0$ there exists a uniquely decodable encoding of K such that

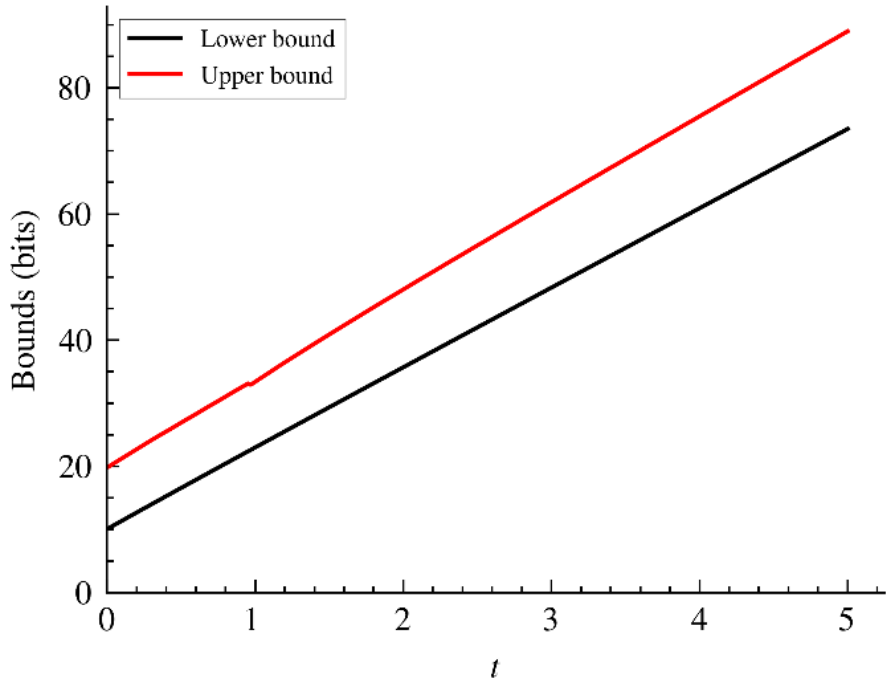
$$L(t) \leq (1+\epsilon)D_{\frac{1+\epsilon(1-\alpha)}{\alpha}}(P||Q) + c(\alpha, \epsilon), \tag{2}$$

with $c(\alpha, \epsilon)$ a constant and $\alpha = \frac{1}{1+t}$.

Gaussian Examples



$P = \mathcal{N}(0, 1)$ and $Q = \mathcal{N}(1, 1)$



$P = \mathcal{N}(0, 1)$ and $Q = \mathcal{N}(5, 1)$

Asymptotic Results

- We want to sample from the product distribution $P^{\otimes n}$ given proposal $Q^{\otimes n}$.

Asymptotic Results

- We want to sample from the product distribution $P^{\otimes n}$ given proposal $Q^{\otimes n}$.
- We can fully characterize the optimal $L(t)/n$ as $n \to \infty$:

Asymptotic Results

- We want to sample from the product distribution $P^{\otimes n}$ given proposal $Q^{\otimes n}$.
- We can fully characterize the optimal $L(t)/n$ as $n \to \infty$:

Theorem 4 For any $t > 0$, let $L_n^*(t)$ be the *minimum Campbell cost* for target $P^{\otimes n}$ and common randomness $\{U_i\}_{i \geq 1} \sim Q^{\otimes n}$. Then, with $\alpha = \frac{1}{1+t}$,

$$\lim_{n \to \infty} \frac{L_n^*(t)}{n} = D_{\frac{1}{\alpha}}(P||Q).$$

Asymptotic Results

- We want to sample from the product distribution $P^{\otimes n}$ given proposal $Q^{\otimes n}$.
- We can fully characterize the optimal $L(t)/n$ as $n \to \infty$:

Theorem 4 For any $t > 0$, let $L_n^*(t)$ be the *minimum Campbell cost* for target $P^{\otimes n}$ and common randomness $\{U_i\}_{i \geq 1} \sim Q^{\otimes n}$. Then, with $\alpha = \frac{1}{1+t}$,

$$\lim_{n \to \infty} \frac{L_n^*(t)}{n} = D_{\frac{1}{\alpha}}(P||Q).$$

This generalizes known results: the *minimum bits/sample* rate is

$$\lim_{n \to \infty} \frac{R_n^*}{n} = D(P||Q).$$

Causal vs. Noncausal Sampling

- A *causal* sampler accepts/rejects each candidate one-at-a-time (K is a stopping time w.r.t. $\{U_i\}_{i\geq 1}$).

Causal vs. Noncausal Sampling

- A *causal* sampler accepts/rejects each candidate one-at-a-time (K is a stopping time w.r.t. $\{U_i\}_{i\geq 1}$).
- Greedy rejection sampling ✓ Poisson functional representation ✗

Causal vs. Noncausal Sampling

- A *causal* sampler accepts/rejects each candidate one-at-a-time (K is a stopping time w.r.t. $\{U_i\}_{i\geq 1}$).
- Greedy rejection sampling ✓ Poisson functional representation ✗
- GRS and the PFR **both achieve** bits/sample rate $D(P||Q)$ as $n \to \infty$.

Causal vs. Noncausal Sampling

- A *causal* sampler accepts/rejects each candidate one-at-a-time (K is a stopping time w.r.t. $\{U_i\}_{i\geq 1}$).
- Greedy rejection sampling ✓ Poisson functional representation ✗
- GRS and the PFR **both achieve** bits/sample rate $D(P||Q)$ as $n \to \infty$.

Theorem 5 For any $t > 0$ let $L_n^*(t)$ be the minimum Campbell cost over *causal* samplers between $P^{\otimes n}$ and $Q^{\otimes n}$. Then, with $\alpha = \frac{1}{1+t}$,

$$\liminf_{n\to\infty} \frac{L_n^*(t)}{n} \geq D_\beta(P||Q), \quad \text{where } \beta = \begin{cases} \frac{\alpha}{2\alpha-1}, & \alpha \in (1/2, 1) \\ \infty, & \alpha \in (0, 1/2]. \end{cases}$$

Causal vs. Noncausal Sampling

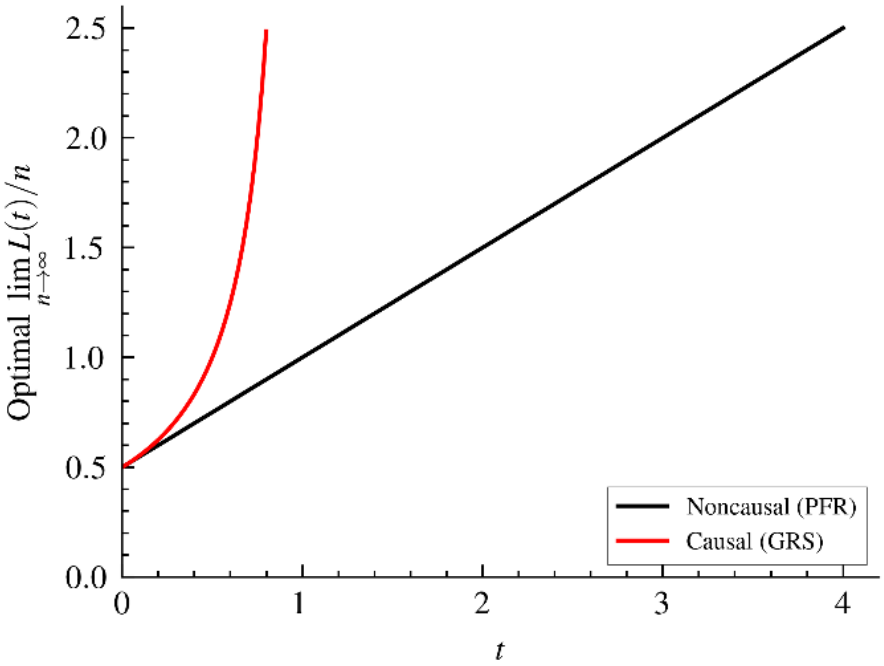
- A *causal* sampler accepts/rejects each candidate one-at-a-time (K is a stopping time w.r.t. $\{U_i\}_{i\geq 1}$).
- Greedy rejection sampling ✓ Poisson functional representation ✗
- GRS and the PFR **both achieve** bits/sample rate $D(P||Q)$ as $n \to \infty$.

Theorem 5 For any $t > 0$ let $L_n^*(t)$ be the minimum Campbell cost over *causal* samplers between $P^{\otimes n}$ and $Q^{\otimes n}$. Then, with $\alpha = \frac{1}{1+t}$,

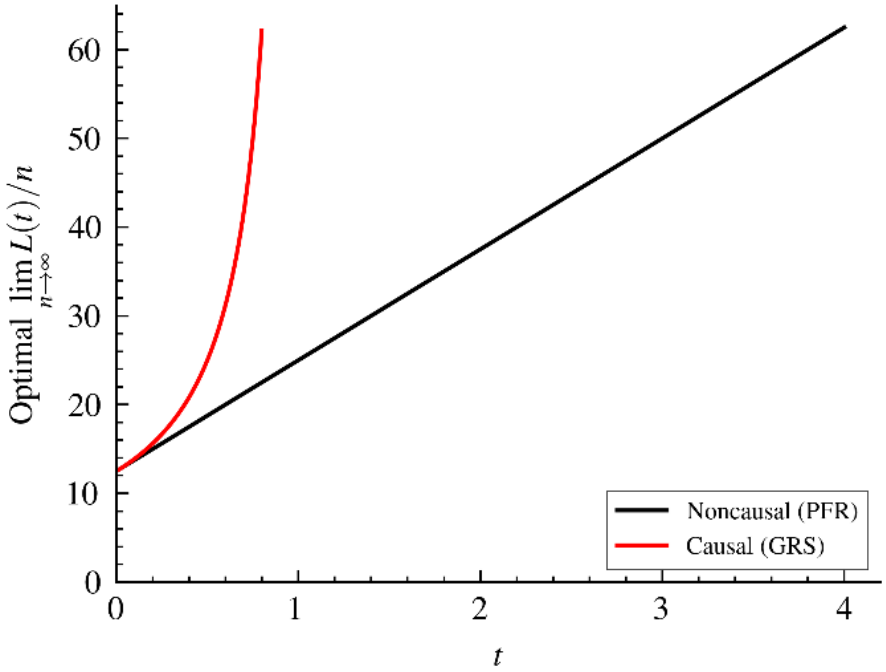
$$\liminf_{n\to\infty} \frac{L_n^*(t)}{n} \geq D_\beta(P||Q), \quad \text{where } \beta = \begin{cases} \frac{\alpha}{2\alpha-1}, & \alpha \in (1/2, 1) \\ \infty, & \alpha \in (0, 1/2]. \end{cases}$$

- $D_\beta(P\|Q) > D_{\frac{1}{\alpha}}(P\|Q)$ in general!!!

Asymptotic Gaussian Examples



$P = \mathcal{N}(0, 1)$ and $Q = \mathcal{N}(1, 1)$



$P = \mathcal{N}(0, 1)$ and $Q = \mathcal{N}(5, 1)$

Greedy rejection sampling does **strictly worse** in the **exponential cost regime**, and the gap is often significant.

Some Open Questions in Channel Simulation

Some Open Questions in Channel Simulation

- *Fast* channel simulation algorithms (linear time complexity in n and $D(P||Q)$), especially for P and Q Gaussians.

Some Open Questions in Channel Simulation

- *Fast* channel simulation algorithms (linear time complexity in n and $D(P||Q)$), especially for P and Q Gaussians.
- Tighter fundamental limits using measures other than the divergence/mutual information.

Some Open Questions in Channel Simulation

- *Fast* channel simulation algorithms (linear time complexity in n and $D(P||Q)$), especially for P and Q Gaussians.
- Tighter fundamental limits using measures other than the divergence/mutual information.
- Practical implementation of common randomness.

Main Takeaways

Main Takeaways

- Machine learning can be used to improve compression, if it is used *in conjunction with* classical techniques.

Main Takeaways

- Machine learning can be used to improve compression, if it is used *in conjunction with* classical techniques.
- Sampling/channel simulation is a practically and theoretically interesting problem.

Main Takeaways

- Machine learning can be used to improve compression, if it is used *in conjunction with* classical techniques.
- Sampling/channel simulation is a practically and theoretically interesting problem.
- The Campbell cost $L(t)$ generalizes the expected message length and can be made more sensitive to the tails of the distribution.

Main Takeaways

- Machine learning can be used to improve compression, if it is used *in conjunction with* classical techniques.
- Sampling/channel simulation is a practically and theoretically interesting problem.
- The Campbell cost $L(t)$ generalizes the expected message length and can be made more sensitive to the tails of the distribution.
- Causal samplers (such as greedy rejection sampling, greedy Poisson rejection sampling, etc.) do **strictly worse** than noncausal samplers in the asymptotic Campbell cost.



Link to arXiv paper!

References

J. Ballé, V. Laparra, and E. P. Simoncelli, "End-to-end optimized image compression," *5th International Conference on Learning Representations*, 2017.

L. L. Campbell, "A coding theorem and Rényi's entropy," *Information and Control*, vol. 8, no. 4, pp. 423–429, 1965.

G. Flamich, M. Havasi, and J. M. Hernández-Lobato, "Compressing images by encoding their latent representations with relative entropy coding," *Advances in Neural Information Processing Systems*, vol. 33, pp. 16 131–16 141, 2020.

P. Harsha, R. Jain, D. McAllester, and J. Radhakrishnan, "The communication complexity of correlation," *IEEE Transactions on Information Theory*, vol. 56, no. 1, pp. 438–449, 2010.

G. Flamich, S. M. Sriramu, and A. B. Wagner, "The redundancy of non-singular channel simulation," *arXiv:2501.14053*, 2025.

E. Lei, H. Hassani, S. S. Bidokhti, "Neural estimation of the rate-distortion function with applications to operational source coding," *IEEE Journal on Selected Areas in Information Theory*, no. 4, pp. 674-686, 2023.

D. Goc and G. Flamich, "On channel simulation with causal rejection samplers," in *IEEE International Symposium on Information Theory (ISIT)*. IEEE, 2024, pp. 1682–1687.

C. T. Li and A. El-Gamal, "Strong functional representation lemma and applications to coding theorems," *IEEE Transactions on Information Theory*, vol. 64, no. 4, pp. 2583–2592, 2018.

References

G. Flamich, S. M. Sriramu, and A. B. Wagner, "The redundancy of non-singular channel simulation," *arXiv preprint arXiv:2501.14053*, 2025.

C. T. Li, *Channel Simulation: Theory and Applications to Lossy Compression and Differential Privacy*. Now Publishers, Inc., 2024, vol. 21, no. 6.

G. Flamich, "Greedy Poisson rejection sampling," *Advances in Neural Information Processing Systems*, vol. 36, pp. 37 089–37 127, 2023.

J. Liu and S. Verdú, "Rejection sampling and noncausal sampling under moment constraints," in *Proc. IEEE International Symposium on Information Theory (ISIT)*, 2018, pp. 1565–1569.

G. Flamich and L. Theis, "Adaptive greedy rejection sampling," in *Proc. IEEE International Symposium on Information Theory (ISIT)*, 2023, pp. 454–459.

C. T. Li and V. Anantharam, "A unified framework for one-shot achievability via the Poisson matching lemma," *IEEE Transactions on Information Theory*, vol. 67, no. 5, pp. 2624–2651, 2021.

S. Hill, F. Alajaji, and T. Linder, "Communication complexity of exact sampling under Rényi information," *arXiv preprint arXiv:2506.12219*, 2025.